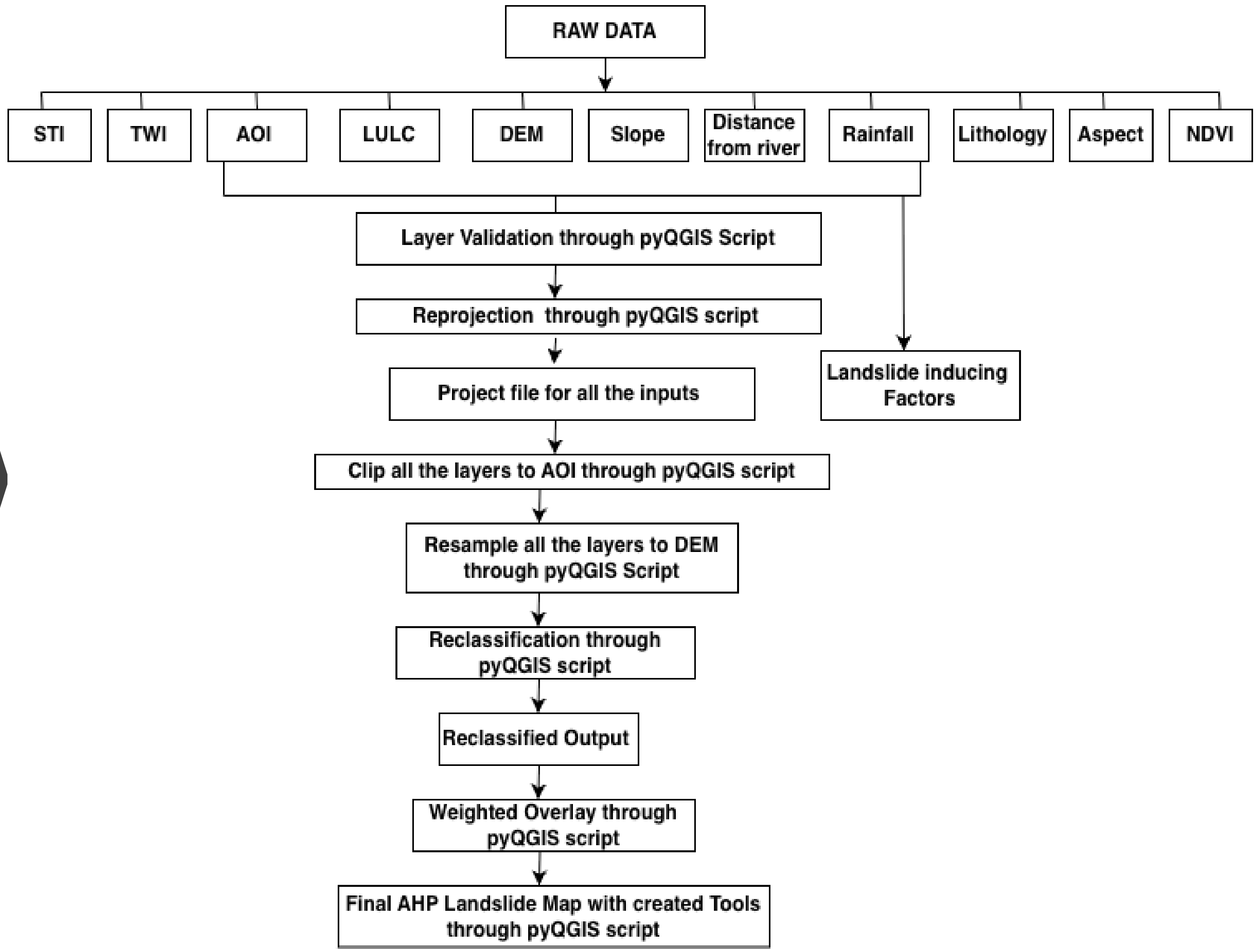
Landslide Susceptibility Mapping for Wayanad District, Kerala, Tool Creation using PyQGIS and Interactive mapping through WEBGIS

Introduction

- **Landslides are one of the most significant natural hazards**, causing loss of life, infrastructure damage, and environmental degradation, particularly in mountainous and hilly regions. Identifying landslide-prone areas is essential for disaster risk reduction and sustainable land-use planning.
- **This project develops a landslide susceptibility map using the Analytical Hierarchy Process (AHP) integrated with GIS techniques.** Multiple landslide conditioning factors, including slope, elevation, aspect, rainfall, lithology, land use/land cover (LULC), distance from rivers, NDVI, STI, TWI, and AOI, are utilized to assess landslide vulnerability.
- **The workflow is automated through PyQGIS scripting**, which performs layer validation, reprojection, clipping, resampling, reclassification, and weighted overlay analysis. Automation minimizes manual effort, improves accuracy, and ensures reproducibility of the landslide mapping process.
- **The final output is an AHP-based landslide susceptibility map and an interactive WebGIS platform**, enabling visualization, analysis, and dissemination of landslide risk information to researchers, planners, and decision-makers for effective hazard management and mitigation planning.
- Objective- **To craete Landslide susceptibility map using the Analytical HierarchyProcess (AHP) integrated tool creation with PYQGIS Script and WEB Mapping.**

Methodology

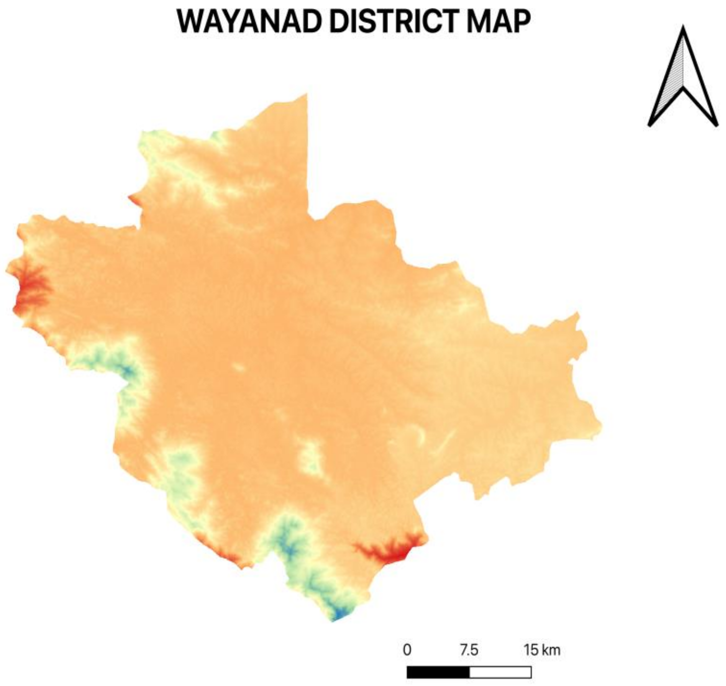


Data Sources

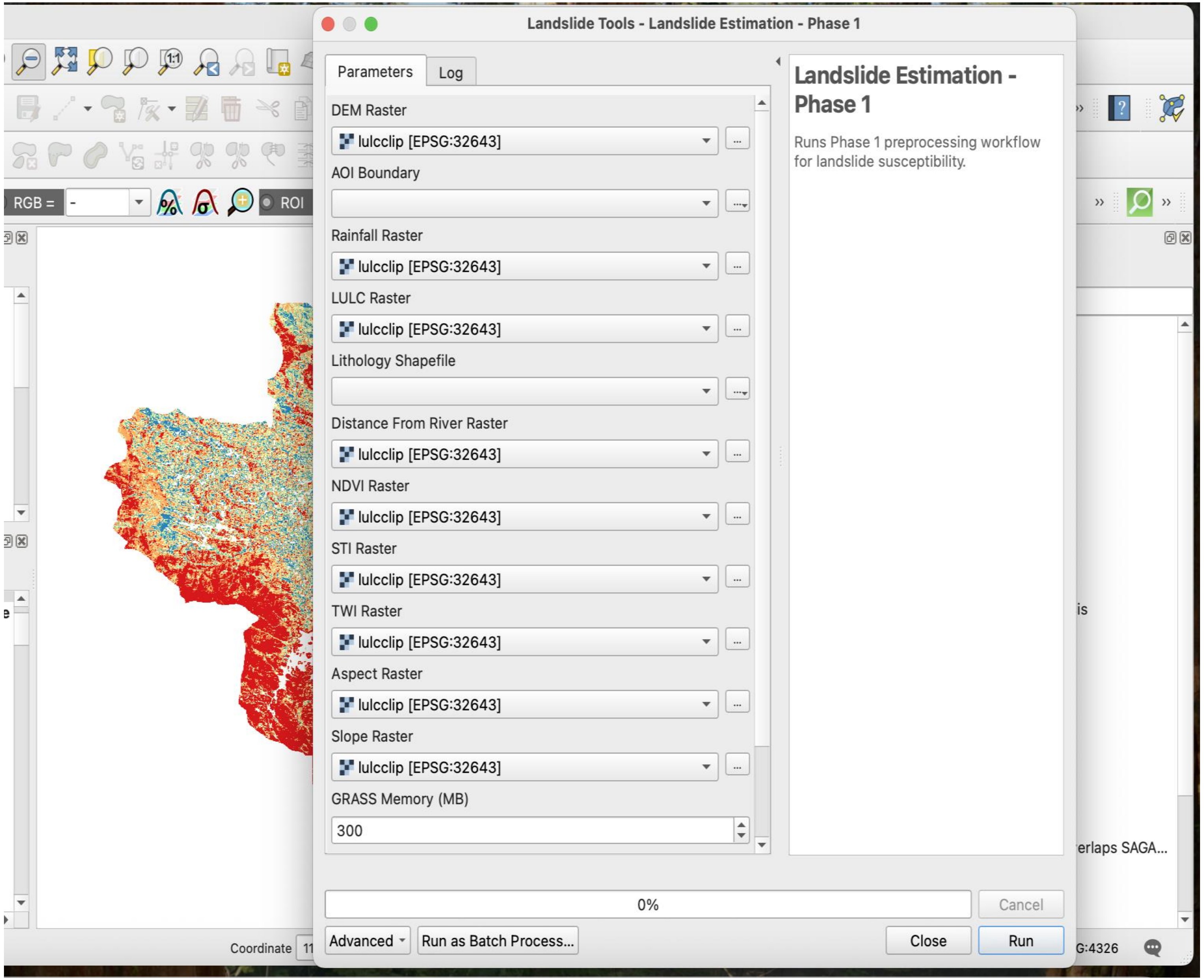
SNO.	Data	Sources
1.	DEM (30M)	USGS Earth Explorer
2.	RAINFALL(0.25x0.25 m)	IMD (India Meteorological Department)
3.	NDVI	GEE(Google Earth Engine)
4.	Lithology	Geological Survey of India
5.	LULC	ESRI

Study Area

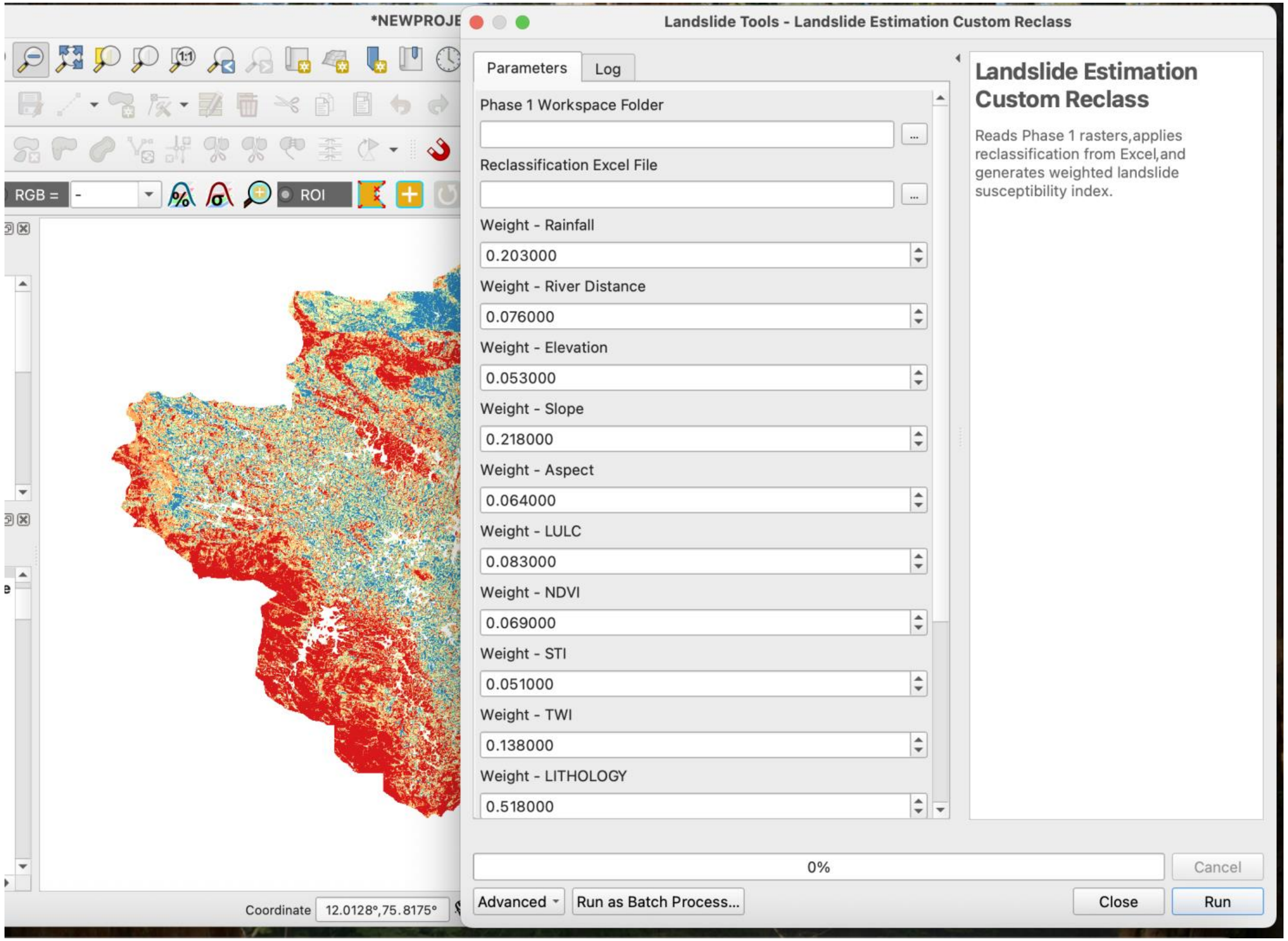
- Wayanad district is located in the north-eastern part of the I
  state of Kerala in the Western Ghats region. It lies bet
  approximately 11°27' and 15°58' North latitude and 75°47' and 7
  East longitude. The district is bordered by Karnataka to the nort
  Tamil Nadu to the east. Wayanad has a mountainous physiog
  with an average elevation ranging from 700 to 2100 metres abov
  level. The region is characterized by rugged hills, deep va
  plateaus, dense forests, and fertile river basins.
- The major landforms of Wayanad include hills, mountain p
  forested uplands, valleys, and river plains. Important peaks su
  Chembra Peak, Banasura Hills, and Brahmagiri Hills form part
  Western Ghats mountain system. The district has rich biodiv
  and extensive forest cover, making it an ecologically significant a
- Several rivers originate from the highlands of Wayanad. The impo
  rivers include the Kabini River and its tributaries such as Panam
  River and Mananthavady River. These rivers provide wate
  agriculture and support the natural ecosystem of the region.
- The soils of Wayanad are mainly forest loam, laterite soil, ar
  loamy soil. These soils are generally fertile and suitable for crops
  as rice, coffee, tea, pepper, cardamom, and rubber. The cool climate, abundant rainfall, and fertile soil make Wayanad an important agricultural district of Kerala.



Tool Created for Landslide Parameter Estimation

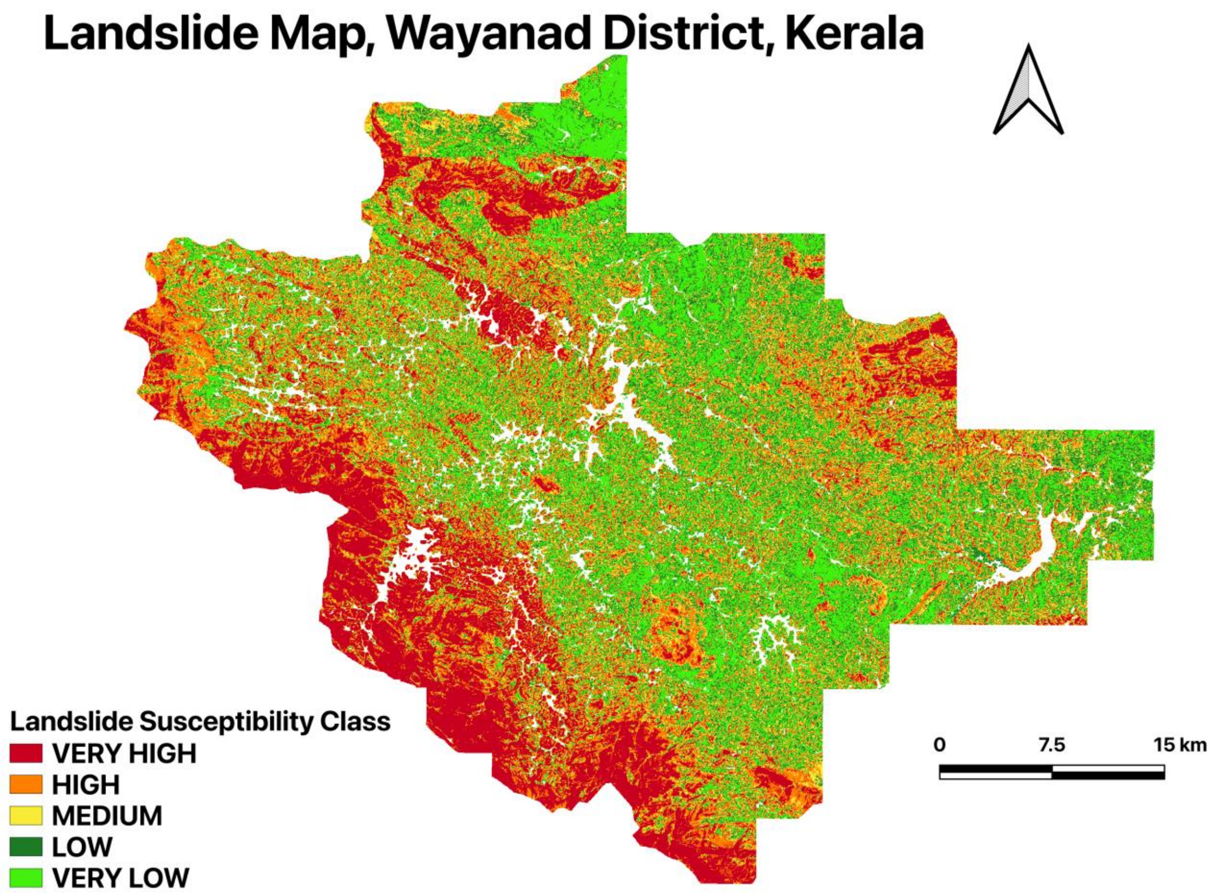


Tool for reclassifying parameters and Final weight assign



Landslide Susceptibility Map

- The landslide susceptibility map of **Wayanad District, Kerala**, was generated using the **Analytical Hierarchy Process (AHP)** integrated with GIS-based multi-criteria analysis to identify areas vulnerable to landslides.
- The map classifies the study area into **five susceptibility zones**: Very Low, Low, Medium, High, and Very High, represented by green to red color gradients.
- The **southern, southwestern, and northwestern regions** exhibit predominantly **high to very high susceptibility**, indicating greater landslide risk due to terrain and environmental conditions.
- Areas shown in **green and light green** represent low-risk zones, while **orange and red zones** indicate regions that require priority attention for disaster mitigation and land-use planning.
- This susceptibility map provides valuable support for **hazard management, infrastructure planning, and risk reduction strategies**, helping authorities identify and monitor potential landslide-prone locations.



All the script Files

https://madhubalapriya2-debug.github.io/Landslide-Scripts/